

PROJECT ONE - Analysis of Student Success Data

BACKGROUND AND OVERVIEW

This project investigates the relationship between psychological stressors and academic outcomes among high school students. Using SQL to merge and analyze two datasets, the study aims to identify at-risk student populations and evaluate the effectiveness of the institutional support systems.

DATA STRUCTURE AND OVERVIEW

The analysis was performed in Google BigQuery using a relational schema consisting of two primary datasets:

student_performance_finalscore		student_performance	
Student_ID	STRING	Student_ID	STRING
Age	INT64	Age	INT64
Gender	STRING	Gender	STRING
Hours_Studied	NUMERIC	Hours_Studied	NUMERIC
Attendance	NUMERIC	Attendance	NUMERIC
Sleep_Hours	NUMERIC	Sleep_Hours	NUMERIC
Stress_Level	NUMERIC	Stress_Level	NUMERIC
Screen_Time	NUMERIC	Screen_Time	NUMERIC
Previous_GPA	NUMERIC	Previous_GPA	NUMERIC
Part_Time_Job	STRING	Part_Time_Job	STRING
Study_Method	STRING	Study_Method	STRING
Diet_Quality	STRING	Diet_Quality	STRING
Internet_Quality	STRING	Internet_Quality	STRING
Extracurricular	STRING	Extracurricular	STRING
Tutoring_Sessions_Per_Week	INT64	Tutoring_Sessions_Per_Week	INT64
Family_Income	STRING	Family_Income	STRING
Exam_Anxiety_Score	NUMERIC	Exam_Anxiety_Score	NUMERIC
Final_Score	STRING	Grade	STRING

Key Technical Operation: Tables were joined via the `Student\_ID` PRIMARY KEY.

EXECUTIVE SUMMARY

This project analyzes the intersection of academic performance and psychological health. Using SQL-driven analysis, I identified heightened anxiety at the "C/D" GPA border as a primary factor in academic failure.

- **Stress as a Variable:** Anxiety accounts for 24% of the variance in final grades ($r^2=0.24$).
- **The Survival Spike:** Students at the C/D threshold report 18.5% higher anxiety than "Safe" peers.
- **Tutoring as a Stabilizer:** Tutoring correlates with a 20.6% score lift and a 25.9% reduction in performance volatility.
- **The Sleep Paradox:** "Poorly Rested" borderline students outperformed "Well Rested" peers but exhibited the highest anxiety scores (6.77), indicating these students are sacrificing their physical and mental well being for temporary academic gains.

INSIGHTS DEEP DIVE

1. The C/D Threshold Spike

The psychological pressure peaks at the point of academic survival.

- "Aspiring D" students show the highest emotional volatility (Sigma 1.87). Once students move into the "Aspiring C" tier, anxiety drops by 12%, suggesting that mitigating the immediate threat of failure is a significant psychological gain for students to achieve.

2. Tutoring as an Emotional Shield

Tutoring provides a dual benefit: raising overall academic performance while stabilizing emotional health.

- Tutoring reduced performance variance by 26%, effectively eliminating "outlier" failing grades. In at-risk groups, it reduced peak anxiety by 23.7%, transforming a high-stakes struggle into a manageable challenge .

3. The Cost of Academic Survival

For borderline students, sleep is a trade-off rather than a simple enhancer.

- Poorly rested students achieved higher scores (64.91) than well-rested peers (60.25) but at the cost of extreme heightened anxiety (6.77). This suggests students are sacrificing their well-being for marginal gains.

STRATEGIC RECOMMENDATIONS

- Prioritize counseling for students in the 1.9–2.1 GPA range to address the 18.5% anxiety spike at the passing threshold.
- Expand access to tutoring as a performance and mental wellness stabilizer to ensure more predictable academic outcomes.
- Encourage 6–8 hours of sleep to break the high-anxiety cramming cycle, prioritizing sustainability over short-term score spikes.

CHALLENGES AND TROUBLE SHOOTING

1. Narrowing Project Scope

The initial data set was vast, offering a wide array of variables and possible correlations. Early on, I recognized the risk of creating a project that was too broad, exploratory and lacking in cohesion so I made the strategic decision to narrow the scope of my analysis to the intersections of “Previous GPA”, “Exam Anxiety” and “Final Scores”. This allowed me to move beyond a surface level exploratory toward building a targeted narrative.

2. Domain Validation and Threshold Calibration

Throughout several iterations of my analysis, I refined my understanding of the `Previous\_GPA` variable. I had initially mapped my “C/D Borderline” values representing the scale for grade points assigned to a class letter grade on a collegiate level (1.5-1.7). However, upon realizing the variable represented cumulative GPA, I recalibrated my threshold to 1.9-2.1 to match the borderline between C and D grades for a cumulative GPA scale in a high school setting. This was a critical adjustment for accuracy, as it reflects the point where cumulative performance triggers institutional interventions like academic probation which is a significant stressor for high school aged students.

3. Quantitative Bridging and Technical Growth

Coming from a design background, I am skilled in deconstructing human patterns and synthesizing complex information into accessible narratives. To bridge the gap into more advanced quantitative analysis, I proactively sought out technical examples of other analytics projects and collaborated with a more experienced data analyst to learn how to apply statistical models to the scope of my analysis. This allowed me to move from basic data extraction to applying Standard Deviation and Linear Regression, demonstrating my ability to rapidly acquire and implement quantitative skills in a technical environment.

REFLECTIONS AND FUTURE SCOPE

While this analysis provides a robust snapshot of academic and emotional stressors, it also opens several avenues for deeper investigation into the student experience:

- I intend to investigate how family income and part-time jobs factor into the relationship between sleep and anxiety.
- With more time, I will analyze the optimal number of tutoring sessions required to improve and stabilize academic outcomes.
- I want to explore mapping exam weights against Letter Grade shifts to identify students most vulnerable to high-stakes testing anxiety and most likely to benefit academically from receiving additional resources.
- As a next step to add an additional layer of depth to this project, I plan to develop my skills in Python and apply its predictive modeling capabilities to identify which students are at risk of psychological stress and poor performance prior to exams.
- With more time and practice utilizing Power BI, I'd want to incorporate visualizations of my data findings to create a clearer and more accessible visual narrative.